

GROUP PROJECT ASSIGNMENT

Software Measurement and Effort Estimation

Formal assignment brief for student group work

Duration	2 weeks	Deliverables	01 Word report and 01 PowerPoint presentation
System type	Existing software or new software	Core focus	GQM/GQIM, size and complexity, COCOMO I/II

1. Overview

In this assignment, each group will conduct a structured software measurement study for a selected software system. The system may be either an existing software product or a newly proposed software system at the stage of requirements, analysis, design, prototype, or early implementation.

The assignment is intended to help students apply software measurement concepts and software effort estimation techniques in a realistic context. Each group is expected to define measurement goals, analyze software size and complexity, and estimate development effort using an appropriate COCOMO model.

2. Group Task

Each group must select one software system and prepare a coherent analysis covering the following three major components:

- 1. Goal-based measurement framework:** Apply GQM or GQIM to define what should be measured and why.
- 2. Size and complexity analysis:** Analyze the selected system using suitable software size measures such as Function Point, Object Point, and Feature Point, and discuss complexity where relevant.
- 3. Effort estimation:** Estimate development effort using COCOMO I or COCOMO II.

3. System Selection

The selected system must be clearly introduced and justified. Groups may choose an existing software product with sufficiently clear specifications, design documents, source code, or feature descriptions, or a newly proposed software project with sufficiently clear requirements, use cases, or design descriptions.

The scope of the system must be realistic, well defined, and manageable within the assignment period.

4. Required Methodology

Each group must use a goal-based approach to justify the analysis. At minimum, the assignment must include a Goal, the main Question or Questions derived from that goal, and the Metric or Metrics used to answer those questions.

Groups may use GQIM instead of GQM if they wish to include indicators in their framework.

5. Required Analysis Components

5.1 GQM / GQIM

- Define at least one clear measurement goal.
- Derive the main questions from that goal.
- Identify the metrics used to answer those questions.

5.2 Size and Complexity

- Analyze the software system using relevant size-related measures. The analysis should include Function Point, Object Point, Feature Point, and a discussion of complexity where appropriate.
- The report should explain why the chosen size measures are suitable for the selected system and how complexity affects development difficulty or implementation effort.

5.3 Effort Estimation

- Estimate development effort using COCOMO I or COCOMO II.
- The report must clearly present the model selected, the assumptions or input values used, the calculation process, and the final estimation result.

6. Required Structure of the Word Report

The Word report should follow the structure below. A separate data collection section is not required for this assignment.

Section	Required Content
1	Introduction and brief presentation of the selected system.
2	GQM / GQIM framework.
3	Size analysis: Function Point, Object Point, Feature Point, and complexity discussion.
4	Effort estimation using COCOMO I or COCOMO II.
5	Discussion of the relationship between measurement goals, size, complexity, and effort estimation.
6	Limitations.
7	Conclusion.

7. Recommended Structure of the PowerPoint Presentation

A recommended presentation structure is: title and group members; system overview; GQM/GQIM; size analysis; complexity discussion; effort estimation using COCOMO I or COCOMO II; discussion; limitations; conclusion.

8. Academic Requirements

- Concepts and formulas must be used correctly.
- Terminology must be clear and consistent.
- Assumptions must be stated clearly.
- The report must be logically organized, and the conclusion must be supported by the analysis presented.

9. Academic Integrity

Students may use software tools to support their work, but the final submission must reflect the group's own understanding and academic responsibility.

Each group is responsible for checking the correctness of formulas and calculations, ensuring consistency across the report, and verifying that the estimation assumptions are reasonable.

10. Final Note

A strong assignment is not the one that includes the largest number of metrics. A strong assignment is the one that clearly demonstrates what is being measured, why it is being measured, how the selected size measures are used, and how they support effort estimation in a meaningful way.